Ethanol Blending in Gasoline – Latvia

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U.S. GRAINS & BIOPRODUCTS COUNCIL

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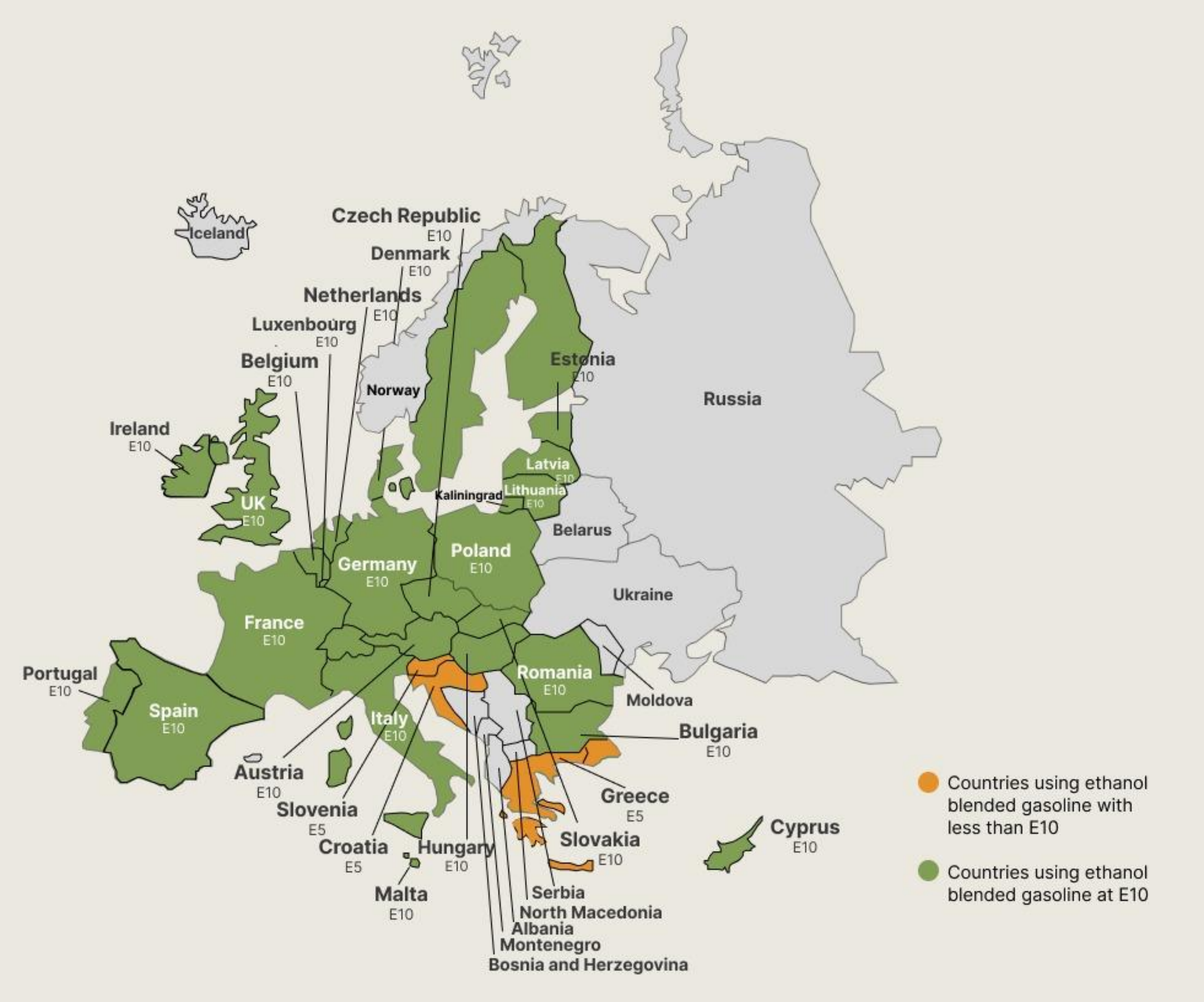
Ethanol Blending in Europe

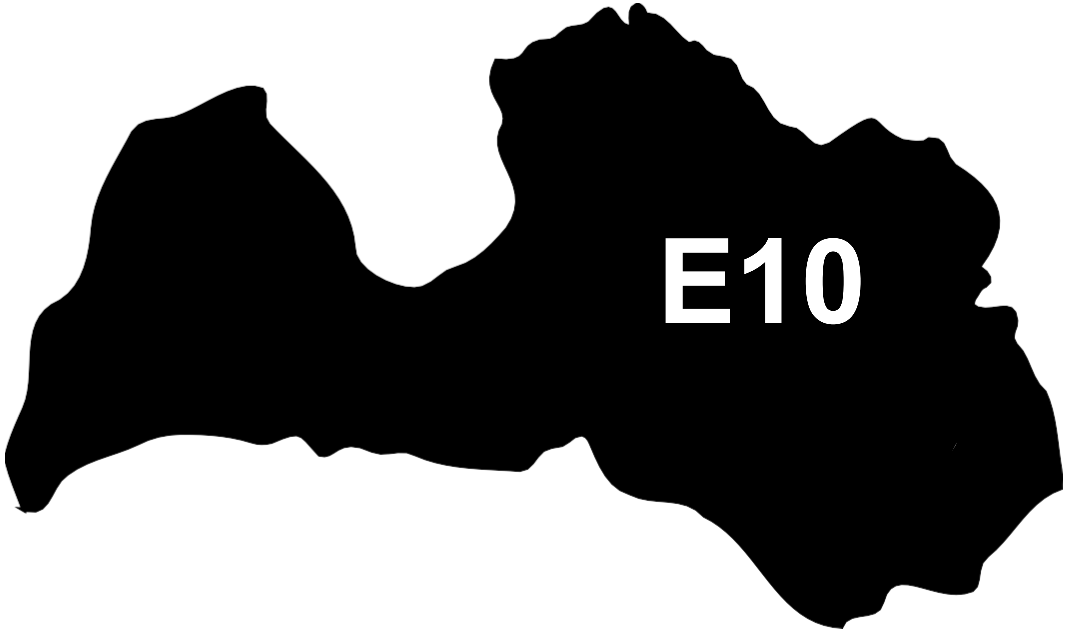
There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Europe to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.



Twenty eight countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

Ethanol Gasoline Blending in Latvia

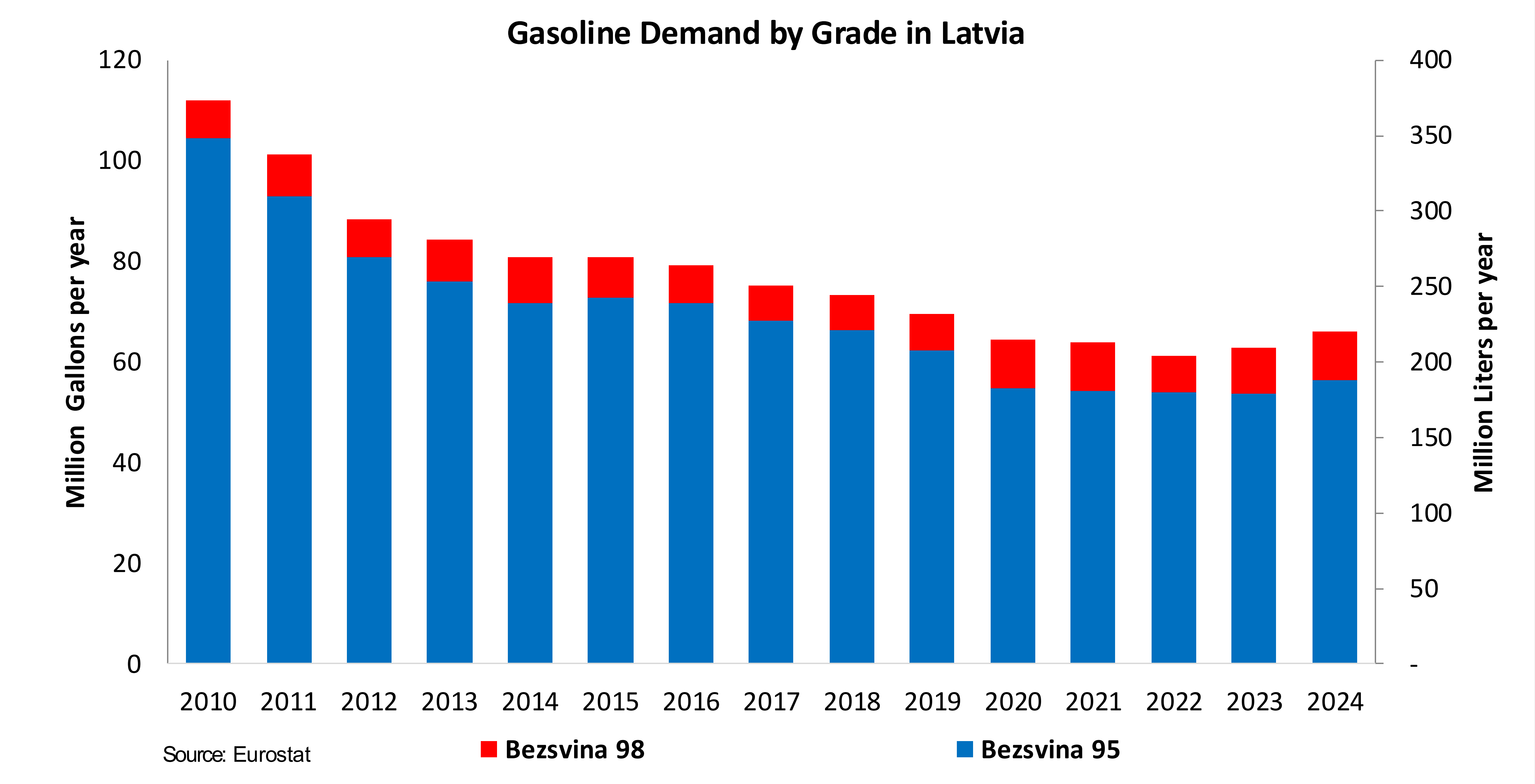


In 2024, gasoline consumption in Latvia was 220 million liters (58 million gallons) and growth rate was -1.1%. There is no gasoline production. Gasoline net imports were 200 million liters (53 million gallons). Latvia complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.4 RON.

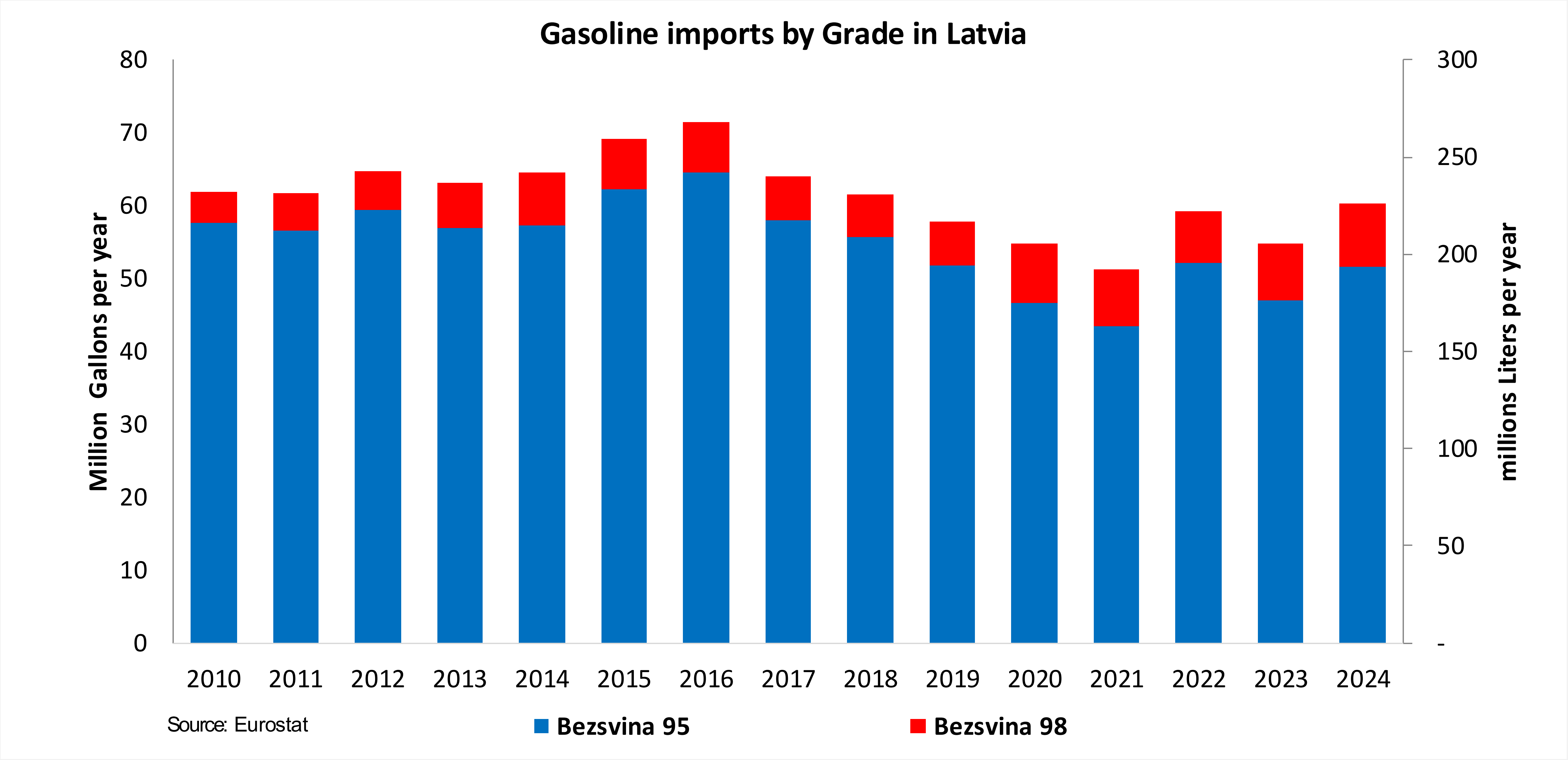
In 2024, ethanol consumption in Latvia was 23 million liters (6 million gallons) representing 10.4% of gasoline demand. Ethanol production and Net Imports were 25 and -3 million liters (6 and -1 million gallons) correspondingly. Ethanol source Cereals 100%.

Source: Eurostat, European Commision

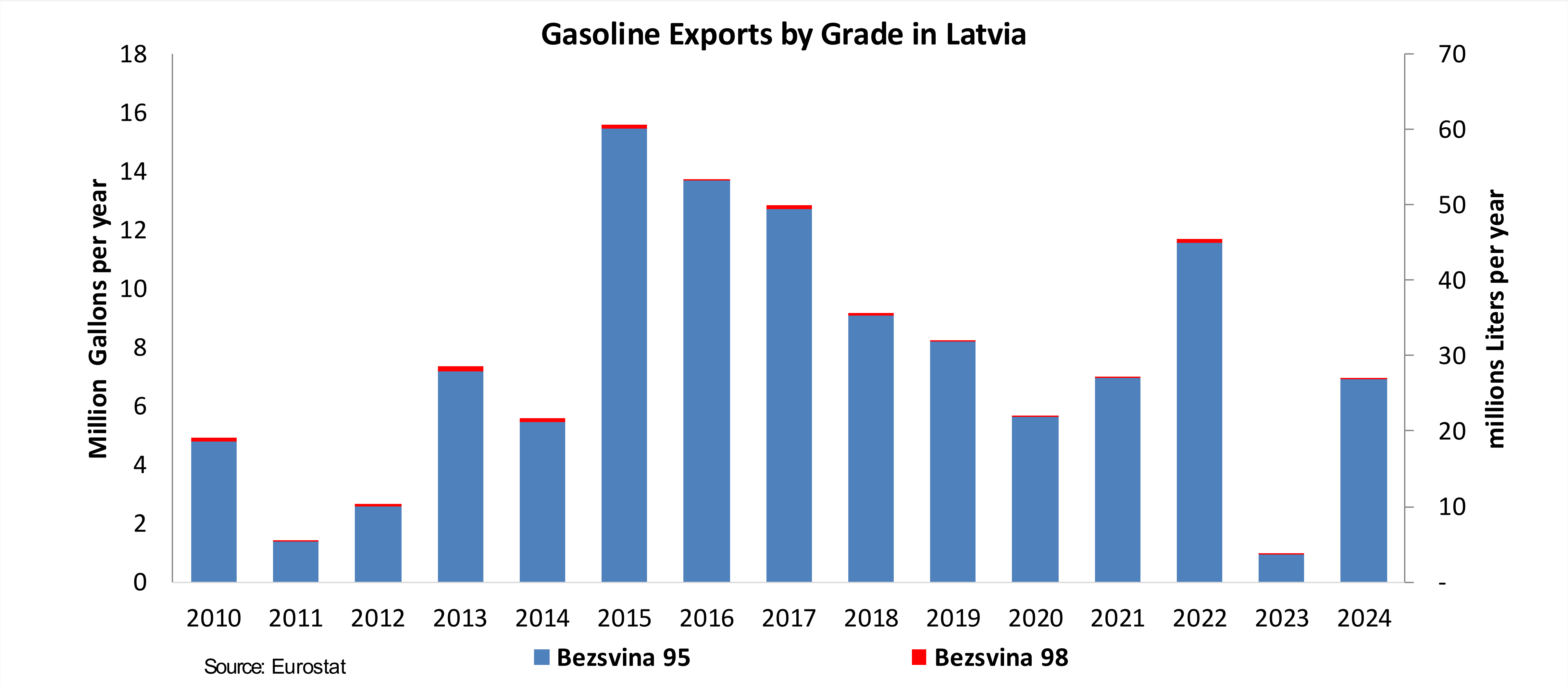
Gasoline Demand in Latvia



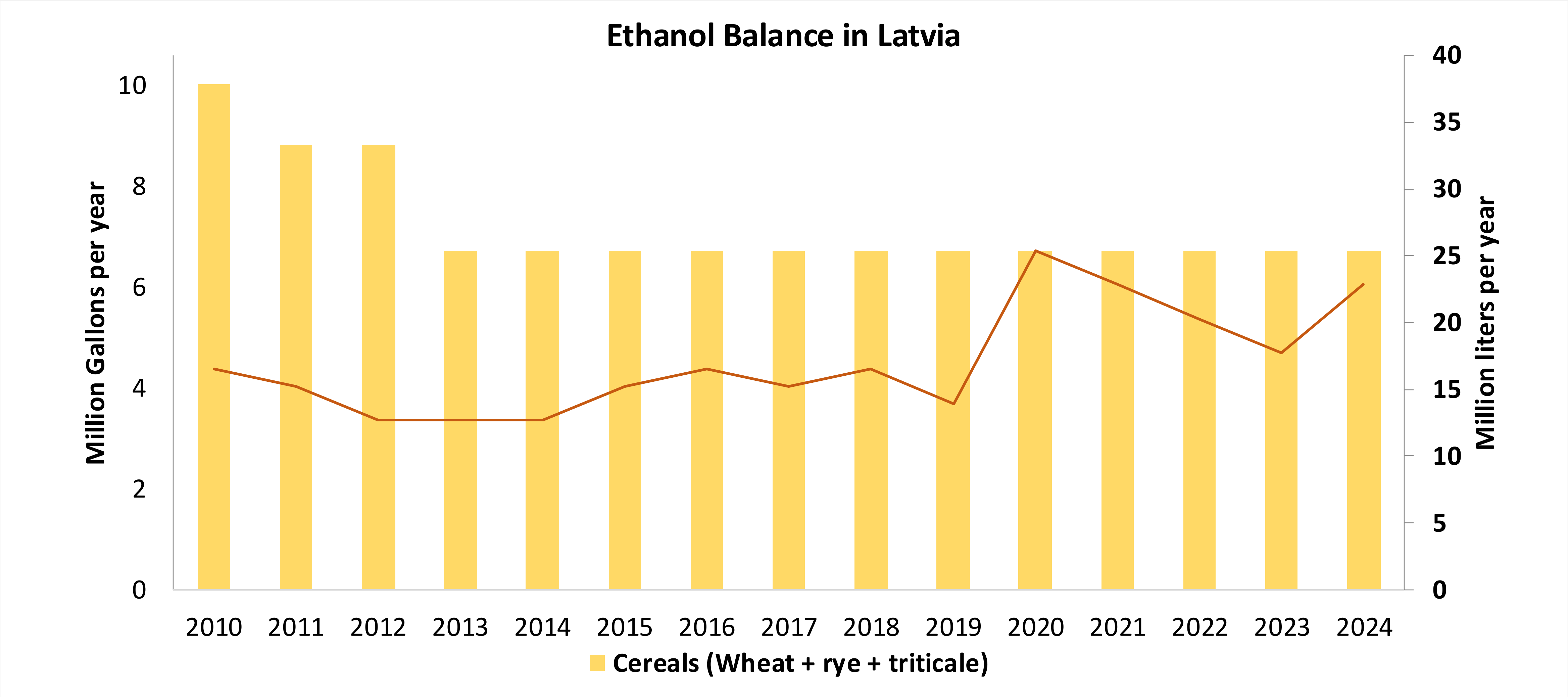
Gasoline Imports in Latvia



Gasoline Exports in Latvia



Ethanol Balance in Latvia



Source: Eurostat, Latvia Statistics

Gasoline Quality in Latvia

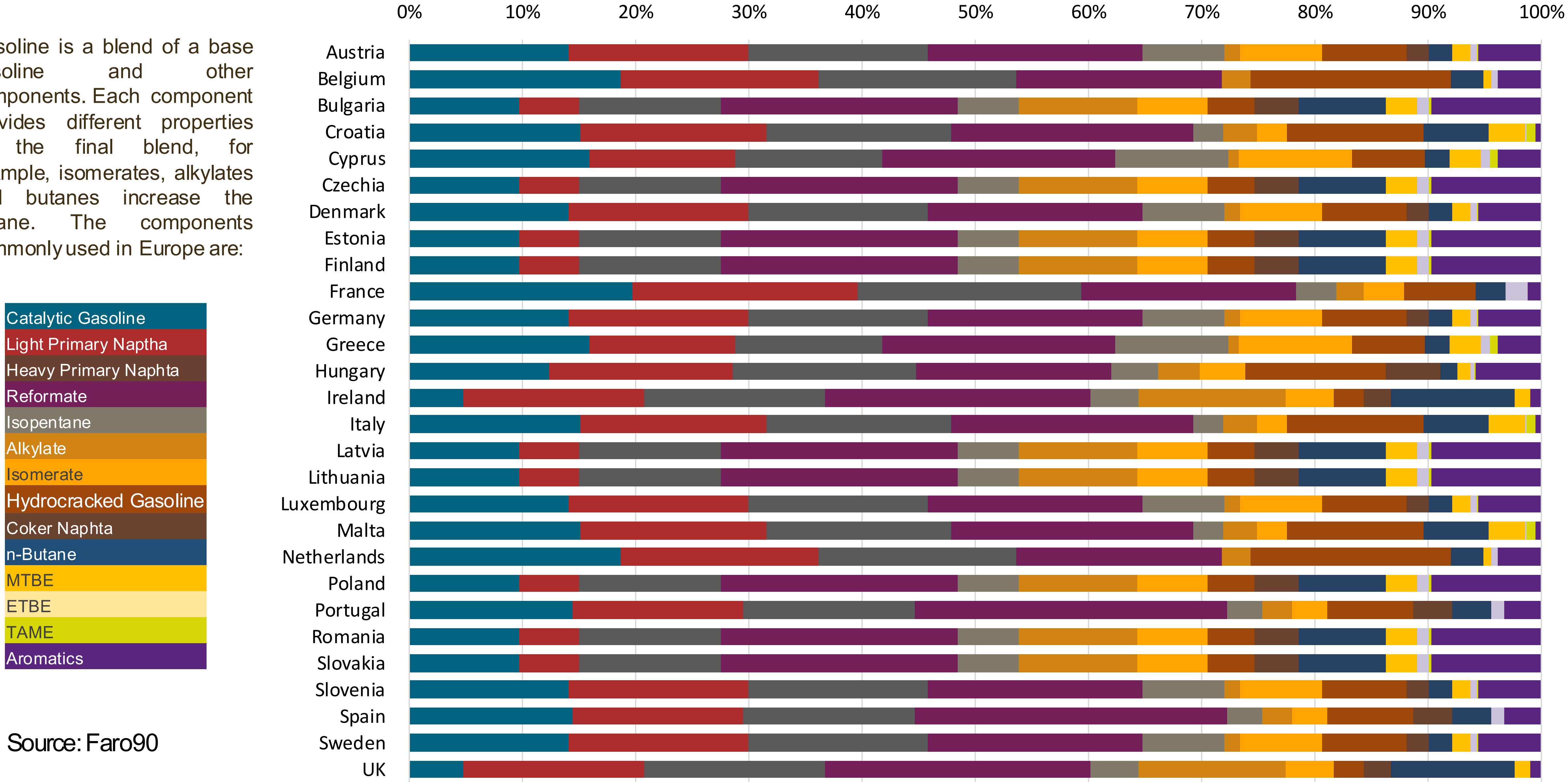
Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	-							
Advanced Biofuels (%cal)	0.2							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

Gasoline Component Blending in Europe

Gasoline is a blend of a base gasoline and other components. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Europe are:



Source: Faro90

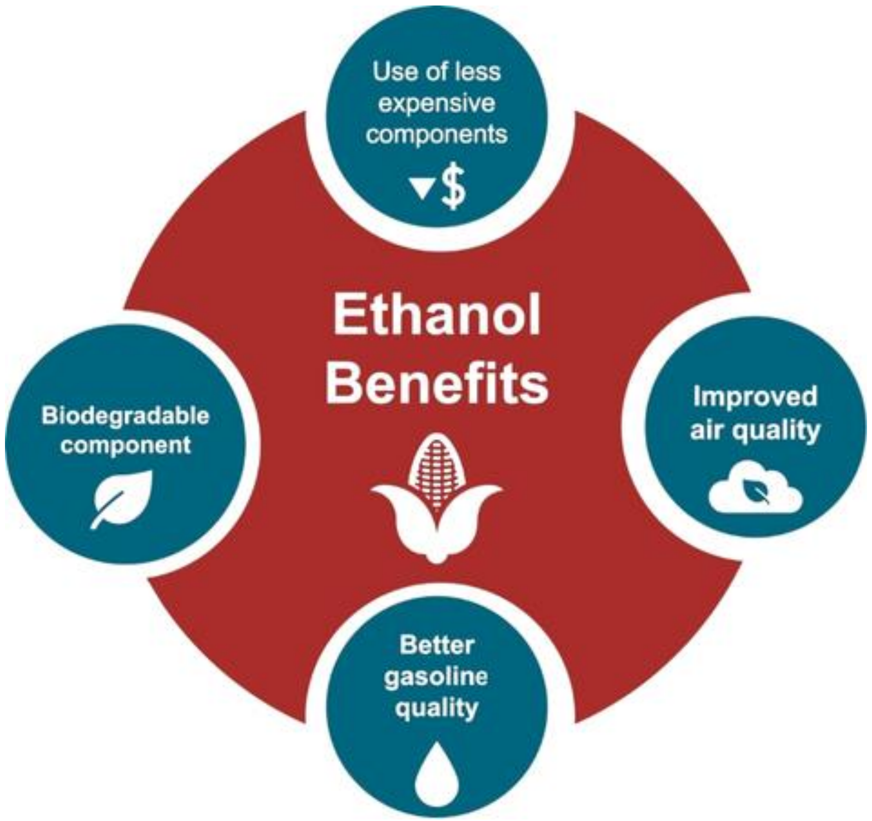
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component.The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives.To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

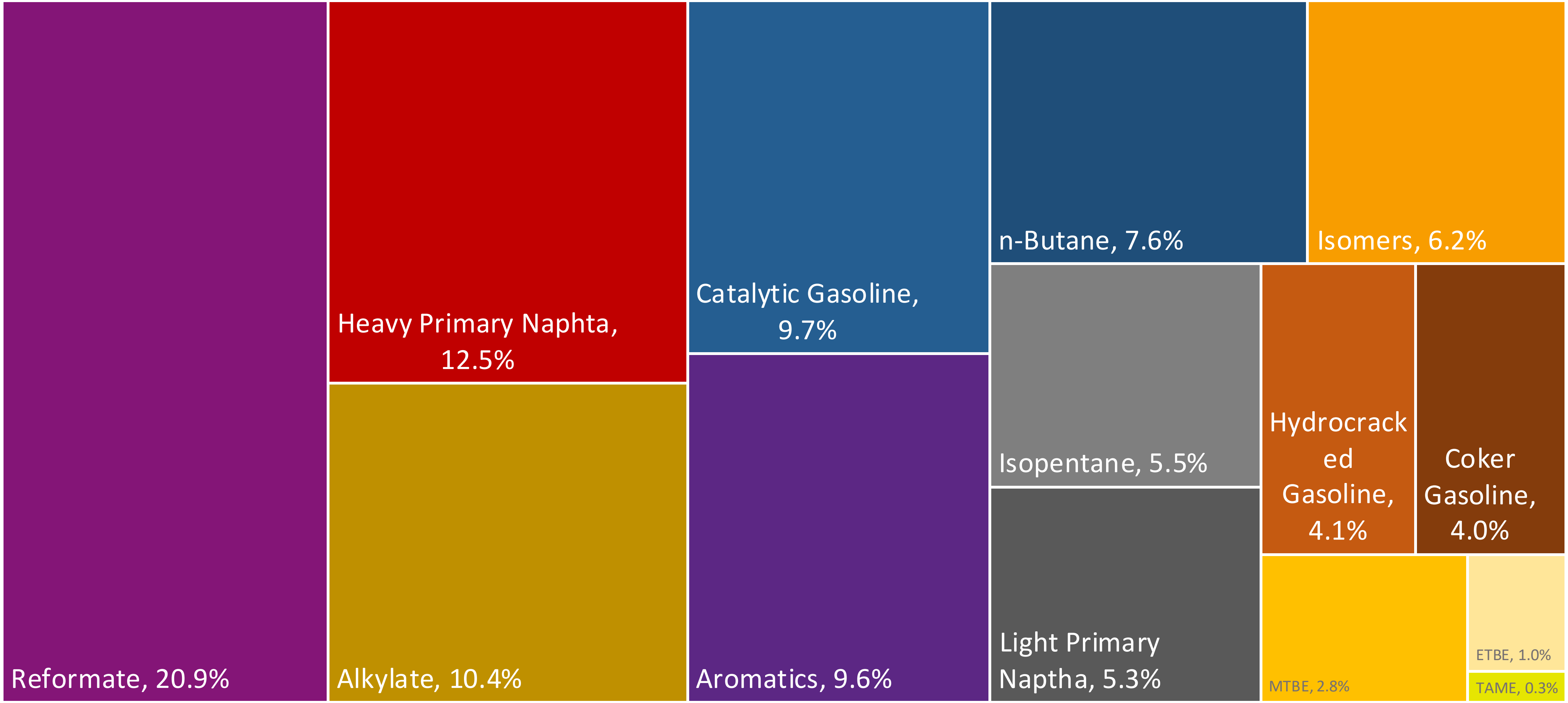
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

The blending model uses gasoline component spot average prices 2024 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.

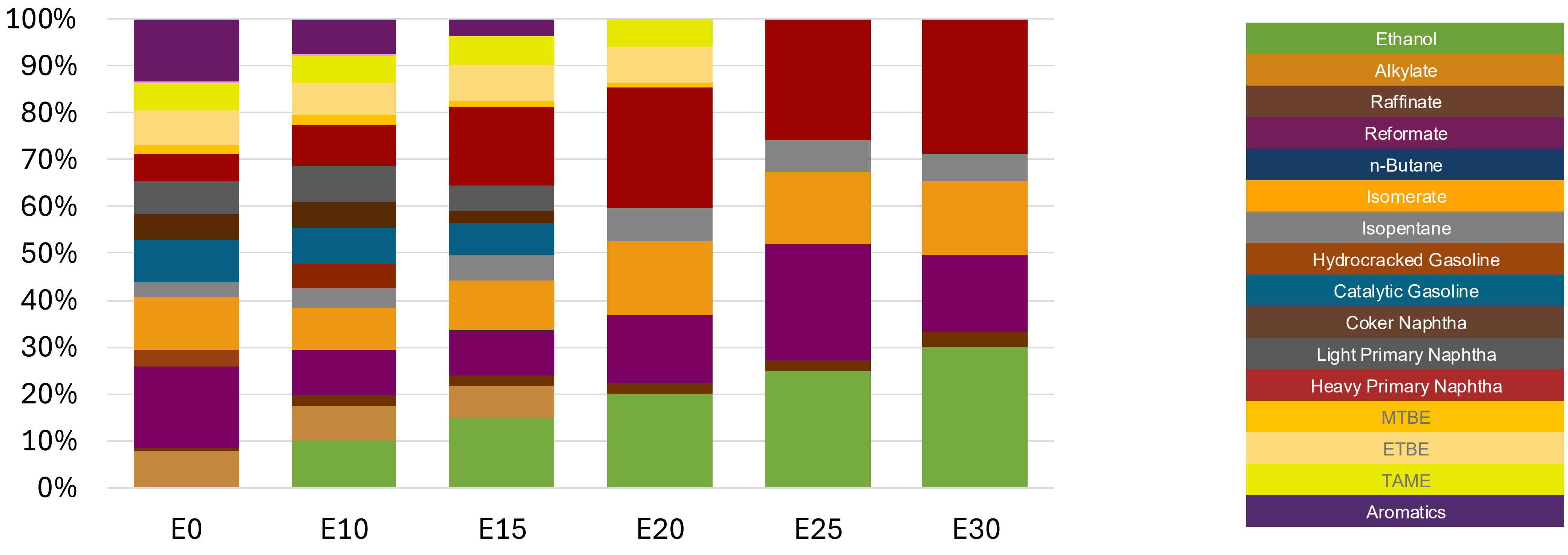


Latvia Available Blending Components



Source: Faro90

Latvia – Gasoline 95 – Constant Octane

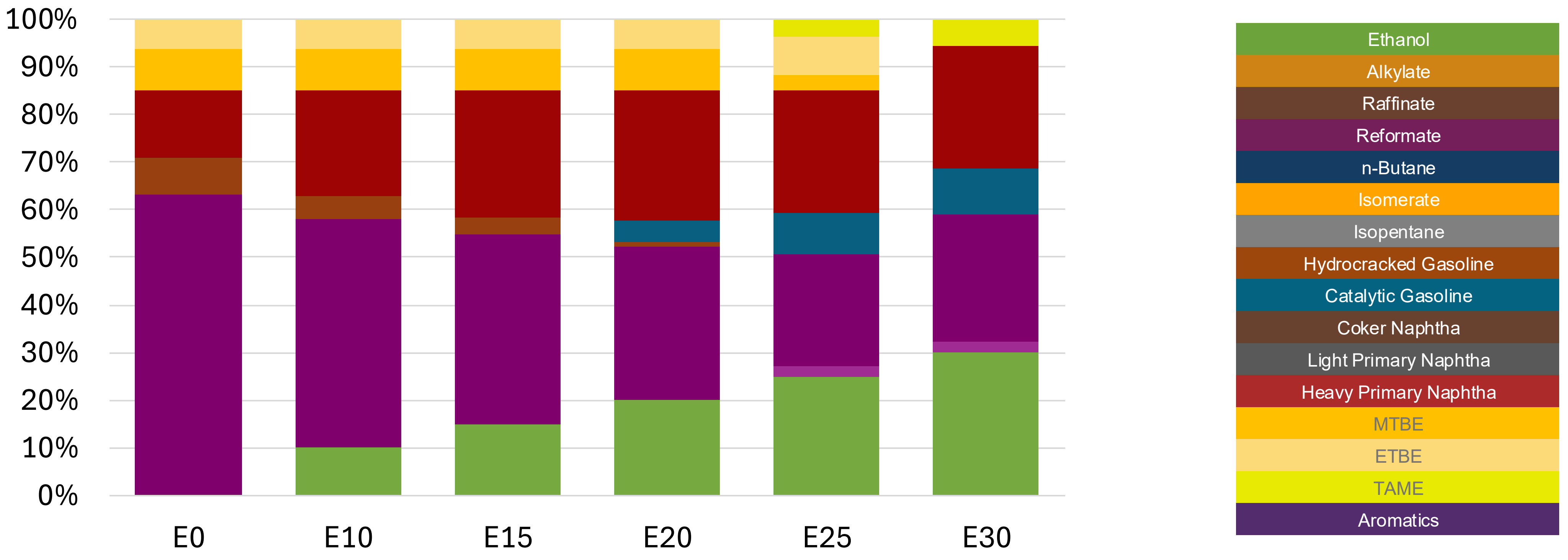


Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.8
Price (US$/gal)	$ 2.274	$ 2.136	$ 2.036	$ 1.921	$ 1.875	$ 1.822

Average CIF 2024 Prices

Source: Faro90

Latvia - Gasoline 98 - Constant Octane

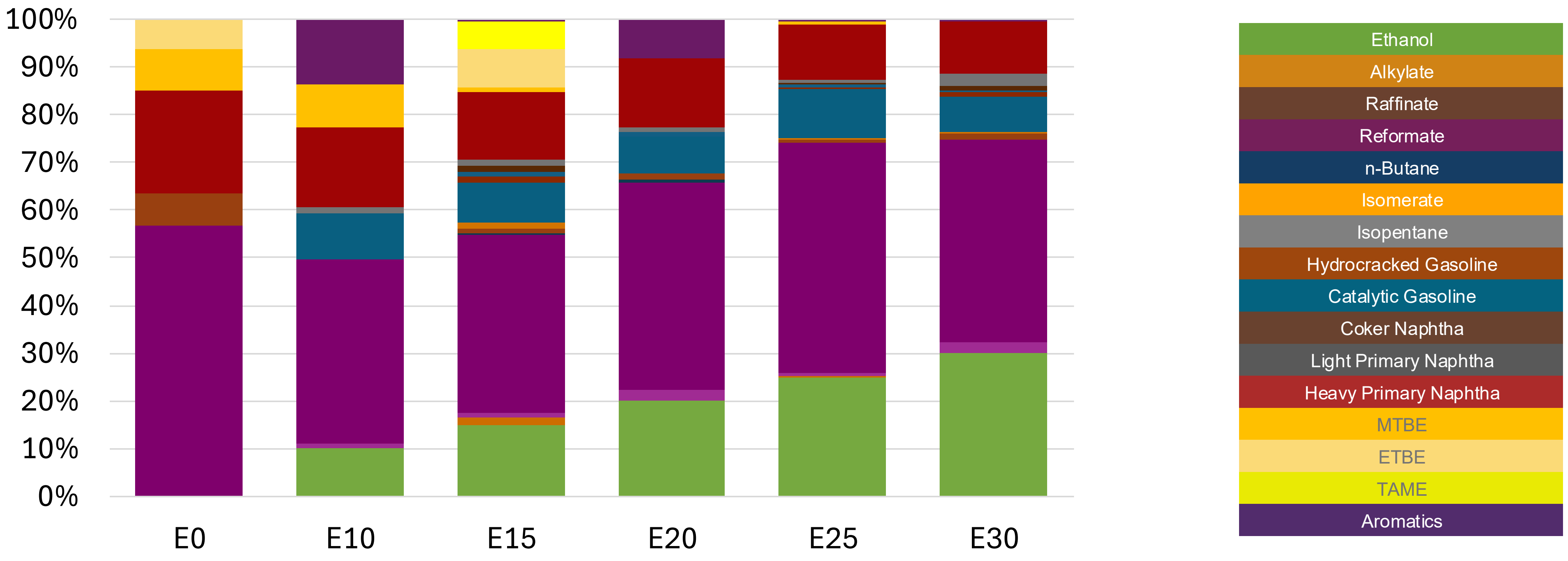


Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US$/gal)	$ 2.294	$ 2.177	$ 2.108	$ 2.036	$ 1.971	$ 1.921

Average CIF 2024 Prices

Source: Faro90

Latvia – Gasoline 95 – Increased Octane

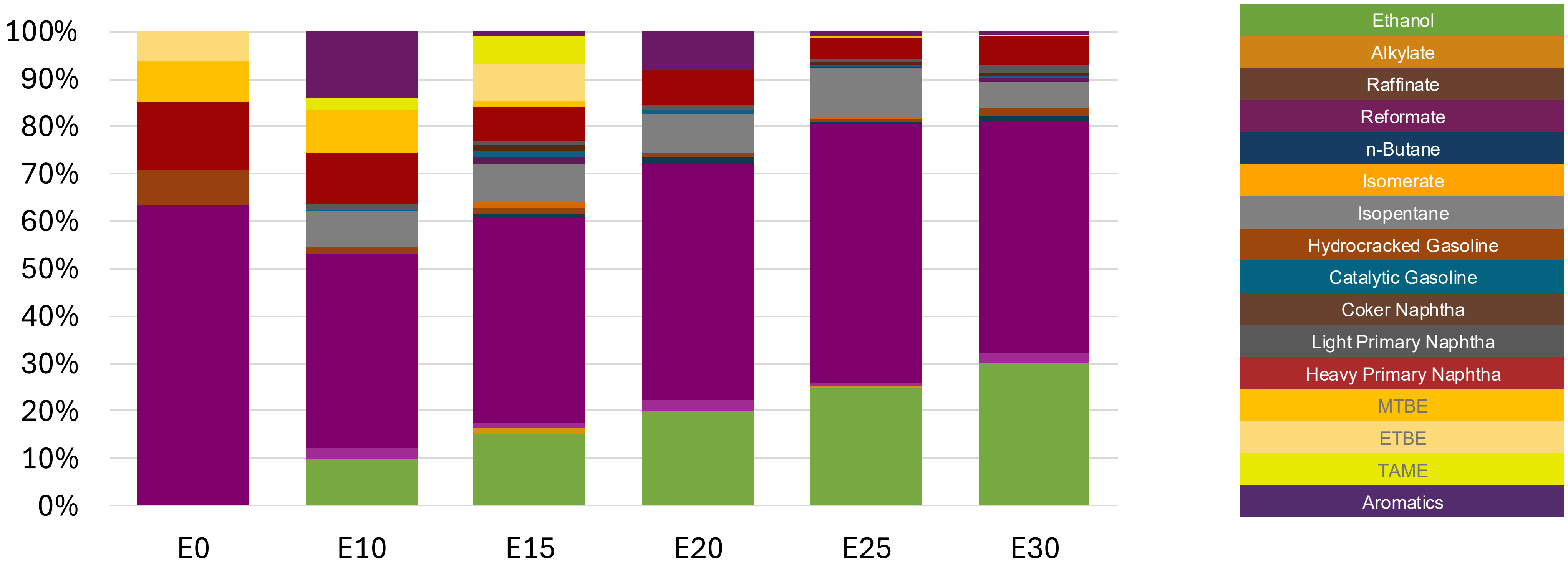


Octane (RON)	95.0	96.2	98.3	98.7	100.5	101.8
Price (US$/gal)	$ 2.162	$ 2.162	$ 2.162	$ 2.162	$ 2.162	$ 2.162

Average CIF 2024 Prices

Source: Faro90

Latvia – Gasoline 98 – Increased Octane



Octane (RON)	98.0	99.0	100.9	101.2	102.8	104.1
Price (US$/gal)	$ 2.294	$ 2.294	$ 2.294	$ 2.294	$ 2.294	$ 2.294

Average CIF 2024 Prices

Source: Faro90

Vehicle Emission Impact for Ethanol Gasoline Blending

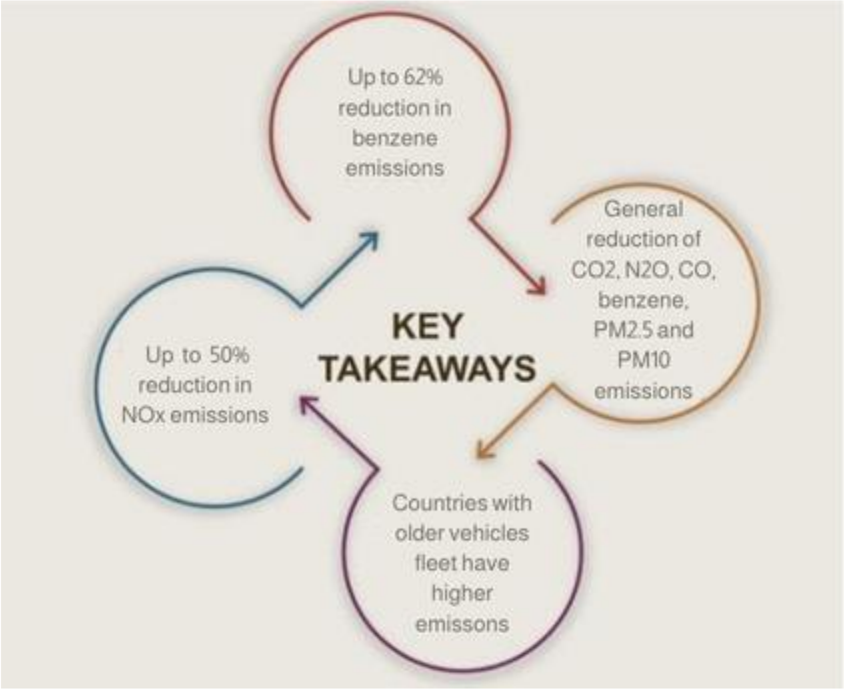
The model used in this analysis takes as a reference the **International Vehicle Emissions Model (IVE).**
The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

Main Results



Source: Bureau of transportation statistics.

Gasoline Vehicle Fleet - Latvia

Type	Weight	Motor	Escape	Recover	Age	No. Vehicles
Auto/Sml Truck	Light	Carburator	3-Ways	PCV Tank	>8 years	-
		Single FI	3-Ways		>8 years	9,291
		Multi FI	3-Ways		>8 years	186,731
			Euro III		>8 years	131,332
			Euro IV		>8 years	216,673
			Euro V		>8 years	69,475
			Euro 6		>8 years	73,328
					4-8 years	43,482
					<4 years	63,509
			Hybrid		All	1,394
		Electric			All	8,366
	Medium	Carburator	3-Ways	PCV Tank	>8 years	-
		Single FI	3-Ways		>8 years	251
		Multi FI	3-Ways		>8 years	19,669
			Euro III		>8 years	25,311
			Euro IV		>8 years	75,052
			Euro V		>8 years	21,387
			Euro 6		>8 years	24,443
					4-8 years	19,668
					<4 years	53,957
			Hybrid		All	1,022
		Electric			All	6,150
Sml Engines	Light	4-Cycle	None	PCV	>8 years	128
			Euro III		>8 years	948
			Euro IV		>8 years	6,679
			Euro V		>8 years	2,223
			Euro 6		>8 years	3,066
					4-8 years	3,297
					<4 years	5,918
			Hybrid		All	52
		Electric			All	313
Truck	Heavy	Carburator	3-Ways	PCV	>8 years	-
		Fuel-Injection	3-Ways		>8 years	13,871
			Euro III		>8 years	24,067
			Euro IV		>8 years	59,352
			Euro V		>8 years	25,855
			Euro 6		>8 years	37,341
					4-8 years	35,094
					<4 years	42,230
			Hybrid		All	10
		Electric			All	469

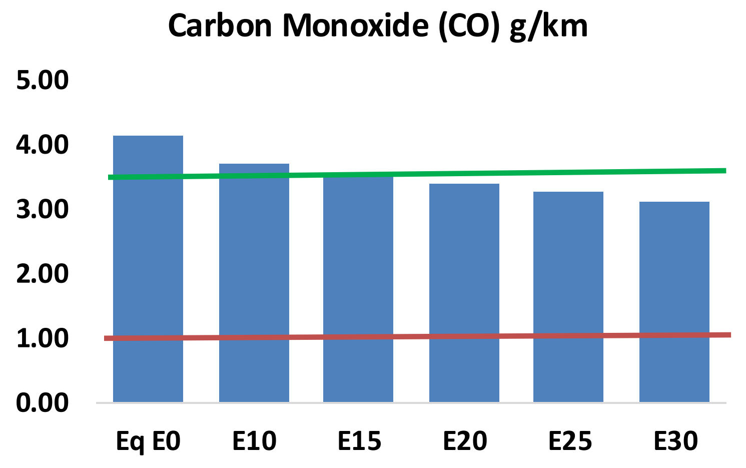
0 50,000 100,000 150,000 200,000 250,000

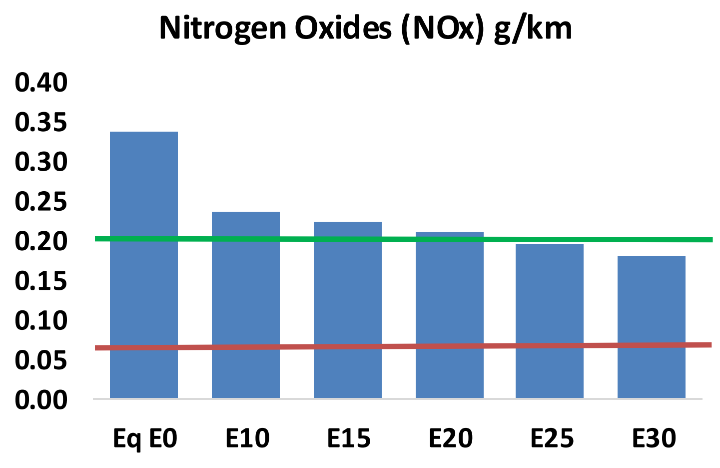
Vehicle Fleet: 1,311,403
Gasoline Vehicle Fleet: 333,918

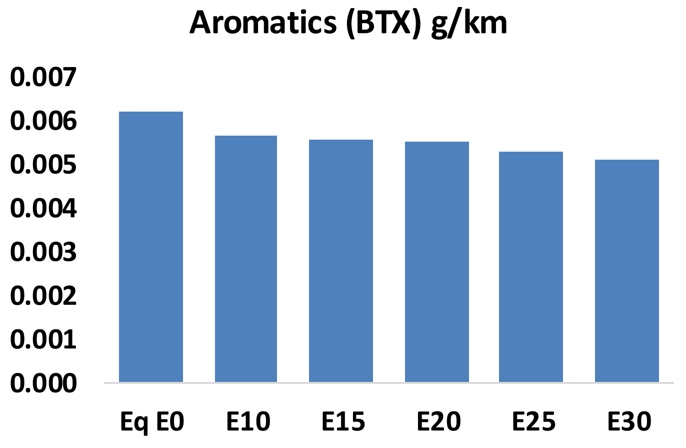
Source: Eurostat, ACEA

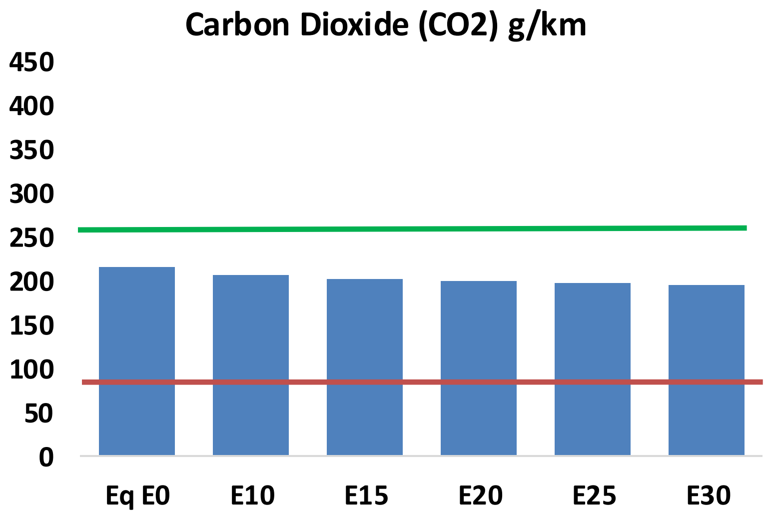
Latvia – Vehicular Emissions

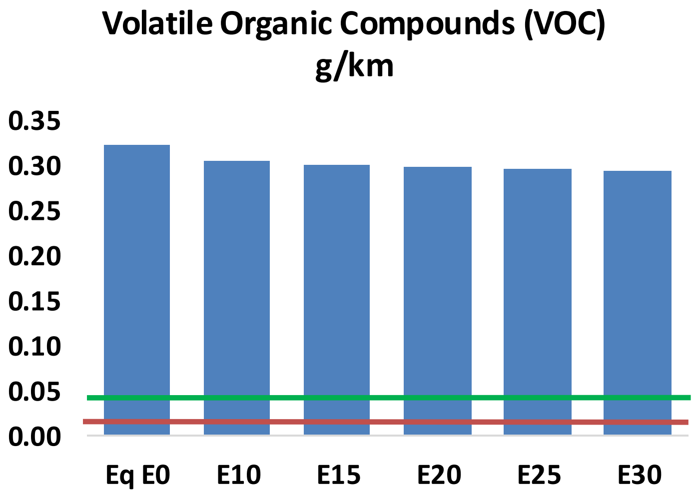
TIER III BIN 125 United States
Euro 6 Standard

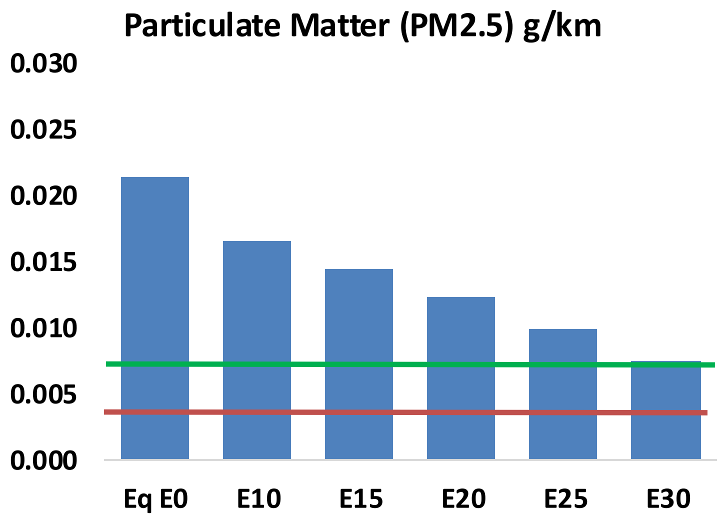












Source: Faro90

Latvia – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	36,253	34,335	32,417	31,014	29,681	28,680	27,419	-11%	-18%
CO2	1,902,185	1,853,272	1,807,076	1,770,744	1,752,766	1,741,942	1,709,833	-5%	-8%
VOC	2,835	2,759	2,682	2,646	2,624	2,610	2,578	-5%	-7%
NOx	2,975	2,232	2,083	1,963	1,851	1,728	1,598	-30%	-38%
SOx	23	21	19	18	16	15	13	-15%	-28%
NH3	639	633	626	629	636	641	646	-2%	0%
N2O	29	29	29	29	30	30	30	-1%	2%
CH4	629	629	629	638	652	664	676	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	35	34	34	33	33	33	33	-4%	-4%
Acetaldehyde	68	78	114	174	237	270	319	68%	249%
Formaldehyde	197	191	223	262	274	303	330	13%	39%
Benzene	55	53	50	49	49	46	45	-9%	-11%
PM2.5	117	104	91	79	67	54	41	-22%	-43%
PM10	71	63	55	48	41	33	25	-22%	-43%

Source: Faro90

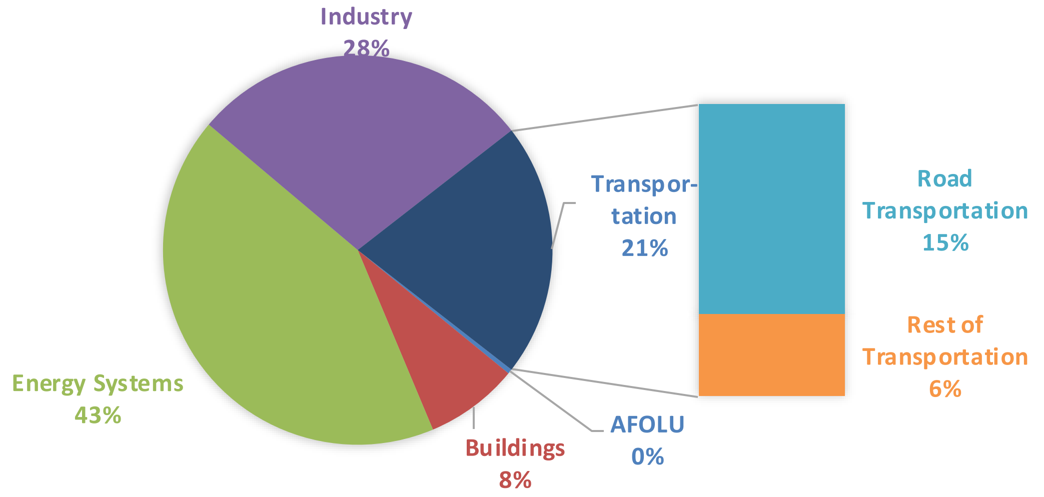
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

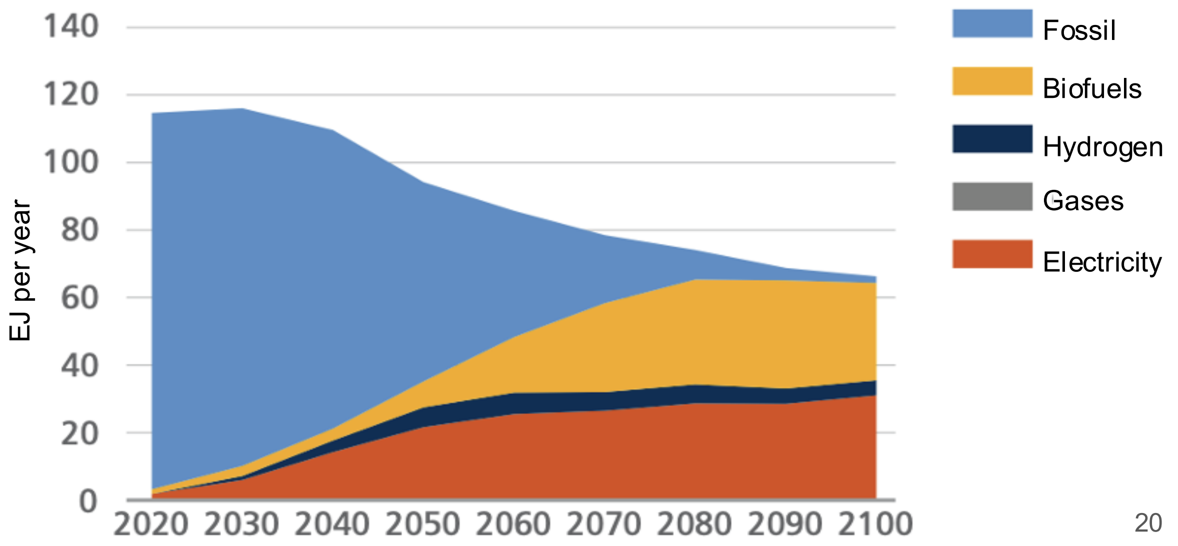
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE).** Results are compared to the current country targets for 2035. Current 2024 country emission are those published by **IPCC**.

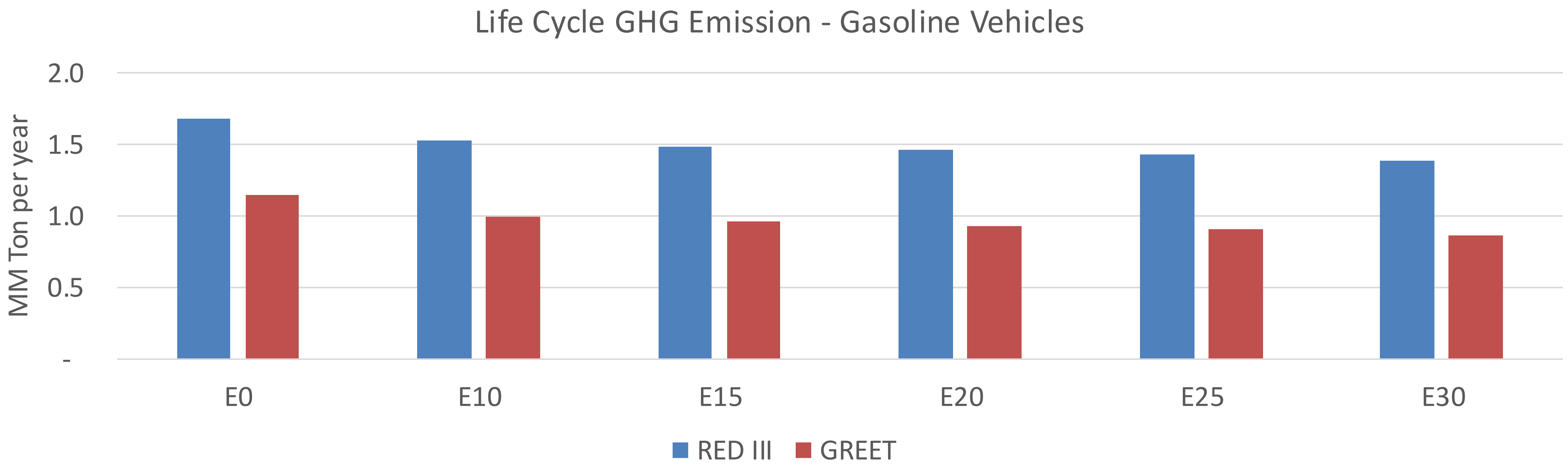
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Latvia – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	12.1
Target @2035 (MMT/year)	6.8
Delta	5.3

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	13.1%	16.2%	18.7%	21.1%	24.3%
RED II/III	0.0%	9.3%	11.6%	13.4%	15.2%	17.5%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	2.8%	3.5%	4.0%	4.6%	5.3%
RED II/III	0.0%	3.0%	3.7%	4.3%	4.8%	5.6%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90